

Solar algae system matches air-purification by 25 trees

The Mushroom World Group has launched a solar-powered, algae-based air-purification system designed to combat urban air pollution. Installed at Swami Vivekananda Park, the technology uses microalgae-driven carbon capture to absorb carbon

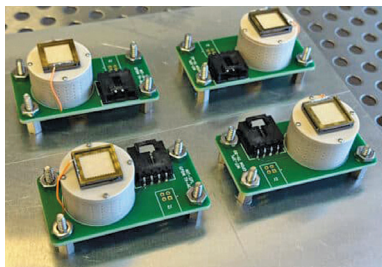


The Mushroom World Group has plans to expand this technology to other cities across India

dioxide, release oxygen, and help reduce particulate matter in densely populated areas. Developed in Bhopal, the compact system targets locations where traditional tree planting is impractical. A single unit can reportedly capture up to 1.5 tons of carbon dioxide annually. The project reflects growing interest in hybrid ecological infrastructure that combines biological processes with engineered environmental solutions for cities.

Satellites may need only one fuel supply

Massachusetts Institute of Technology (MIT) engineers have demonstrated a dual-mode propulsion system that could make small satellites faster, more agile, and fuel-efficient by using a single propellant for both chemical and electric



These four flight unit electro spray thrusters were delivered by MIT Space Propulsion Laboratory to NASA for the upcoming Green Propulsion Dual Mode (GPDM) mission (Image Credit: Amelia Bruno)

thrusters. The technology combines the high-thrust performance of chemical propulsion with the precision and efficiency of electrospray thrusters, eliminating the need for separate fuel systems. Developed at MIT's Space Propulsion Laboratory in Cambridge, the system

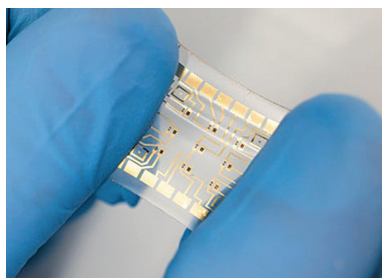
uses the ASCENT green monopropellant, originally designed for chemical propulsion, to power both thruster types. Tests showed that the fuel delivers electric-thruster performance comparable to that of conventional ionic propellants. The innovation could enable compact CubeSats to undertake more complex missions, including deep-space exploration.

GaN chips shrink power systems

Researchers at the US Department of Energy's Oak Ridge National Laboratory in Tennessee have developed gallium nitride (GaN)-based power converters that are smaller, lighter, and more energy-efficient than conventional silicon designs, potentially reducing costs for next-generation AI data centres. The new converters use GaN semiconductors that switch 10 to 20 times faster than silicon while minimising energy losses. The technology was built and validated at the Grid Research Innovation and Development Centre (GRID-C). Tests showed the compact converters can lower installation and maintenance costs while offering greater deployment flexibility. The advancement addresses growing demand for efficient power systems in AI infrastructure.

Stretchable patch detects heart failure risks

Researchers at the University of Chicago's Pritzker School of Molecular Engineering in Chicago have developed an AI-powered stretchable computing patch capable of processing



The new device, designed and tested in collaboration with researchers at Argonne National Laboratory, was made possible by the development of manufacturing processes that allow organic electrochemical transistors to be printed onto flexible surfaces (Photo by John Zich)

health data directly on the body in milliseconds, eliminating the need for external servers or wireless connections. The skin-like device uses large-scale arrays of organic electrochemical transistors to run AI algorithms locally, reducing latency, power consumption, and privacy risks. The patch,

engineered, achieved 99.6% accuracy in tracking cardiac electrical wavefronts linked to ventricular fibrillation. Researchers say the technology could enable smarter wearable and implantable medical devices for real-time health monitoring and intervention.

Lab chip reveals hidden drivers of gut disease

Developed by a team at Harvard University's Wyss Institute for Biologically Inspired Engineering in Boston, the model identified stromal fibroblasts as major drivers of inflammation and tissue damage. Researchers have created the most comprehensive inflammatory bowel disease (IBD)-on-a-chip model to date, providing a new platform for studying disease progression, fibrosis, cancer risk, and treatment responses. The human colon chip replicates key features of IBD using patient-derived intestinal and stromal cells.